

Algebraic Jacobian

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About This Blueprint

This is the authoritative route specification for the Algebraic Jacobian Challenge Rebuild. It fixes one construction: represented degree- g divisors, the rank-one Abel isomorphism, translated rank-one charts over a separably closed field, finite-stage descent of their glued carrier and universal class, finite-Galois descent, and the same-carrier handoff to the challenge interfaces. A formalizer following this text has no remaining route choice.

Each numbered mathematical package states its objects and quantifiers, records the exact public Lean handoff, and visibly classifies its state as *proved*, *imported*, *conditional*, *planned*, or *unresolved*. A `\leanok` marker means that every attached declaration is checked and reachable from the public `AlgebraicJacobian` root. It may certify a conditional implication; the displayed antecedents then remain hypotheses, and the marker gives no completion credit to the corresponding route instantiation. Textual declarations outside that root are named only as unverified implementation artifacts and receive no `\leanok` marker.

Chapter 1

Route contract and hypothesis ledger

1.1 Fixed objects and functors

Definition 1 (The AJCR curve and its degree-zero Picard functor). Let K be a field and let

$$C \longrightarrow \operatorname{Spec} K$$

be smooth of relative dimension one, proper, and geometrically irreducible. Put $g = g(C) = \dim_K H^1(C, \mathcal{O}_C)$. For a K -scheme T , write

$$\operatorname{Pic}_C^0(T) = \operatorname{pic0TypeFunctor}(C)(T)$$

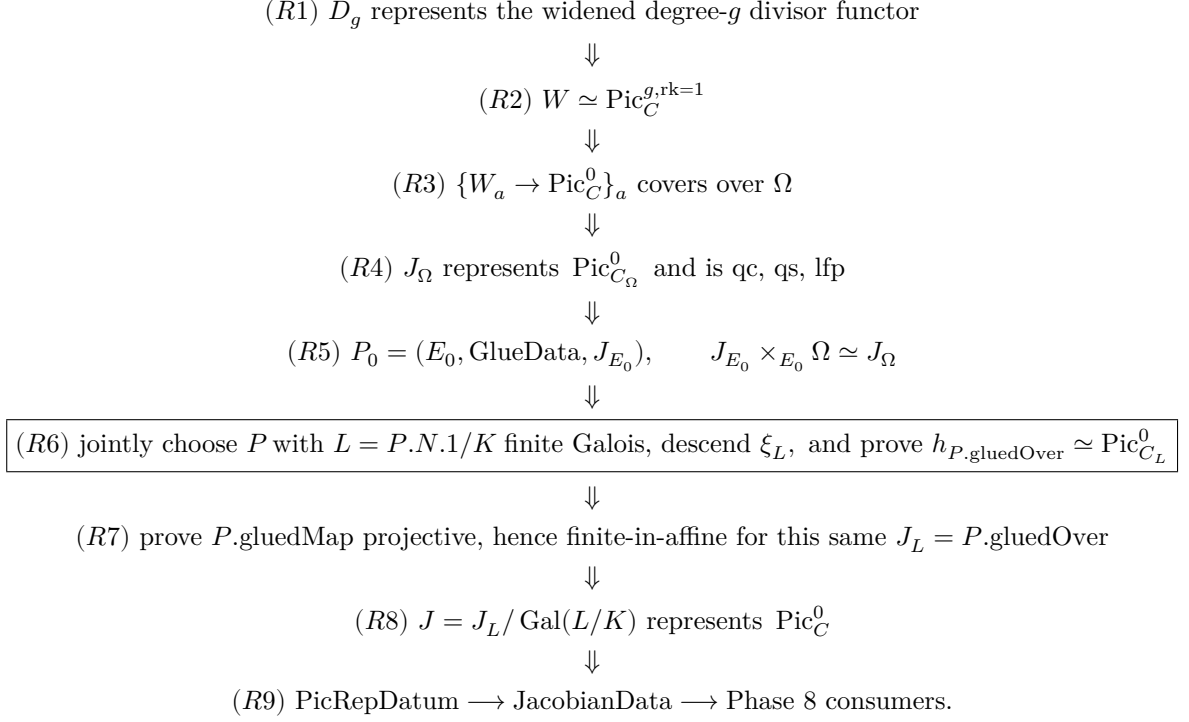
for the degree-zero subgroup of the big-etale-sheafified relative Picard functor used in Lean. Its elements are classes whose degree after every field-valued pullback is zero. This notation does *not* assert that this functor has already been identified with the identity component of a Picard scheme.

No rational point of C is assumed. In particular, no step may introduce a global K -point merely to normalize a divisor or a Picard class.

Proof. This is the exact binder of the public Lean interfaces: $[\operatorname{Field} K]$, $C : \operatorname{Over}(\operatorname{Spec} K)$, and the three displayed geometric typeclasses. The degree-zero functor is defined before any representer and is therefore the object which every later `RepresentableBy` certificate must represent. $\square$

1.2 The one permitted route

The dependency order is fixed as follows. Every arrow is a mandatory dependency, not a status claim. The boxed break marks the first missing mathematical producer; later arrows describe blocked consumers, not a change of route.



The route does not pass through a quotient of the unrestricted admissible-degree Abel map, a symmetric-power/Weil construction, an Albanese first construction, or a filtered-colimit theorem. Those branches are quarantined in [Appendix 38](#).

1.3 Hypothesis ledger

Fix the canonical separable closure $\Omega = \text{SeparableClosure } K$. Thus Ω/K is algebraic, separable, and normal, and Ω is separably closed. The exact Lean instantiation carries $\text{Algebra.IsSeparable } K \ \Omega$ and the normality used by the normal-closure construction. Every finite coefficient set used in R6 must first lie in a finite subextension $E/K \subset \Omega/K$ carrying both $\text{Module.Finite}_K E$ and $\text{Algebra.IsSeparable } KE$. A generic `GluePackage`, however, is constructed under algebraicity alone and does not itself produce the latter certificate. The checked carrier comparison may therefore first be built at an arbitrary algebraic finite stage E_0/K . The missing universal-class producer must separately establish separability, enlarge all finitely many coefficients to a finite normal subextension inside Ω , and only then finalize the legacy package P . Accordingly its literal final field $L = P.N.1$ is already finite Galois over K ; no later step changes the carrier. The following ledger is exhaustive.

ID	Hypothesis	Scope and discharge
H1	K is a field.	Global; never discharged.
H2	C/K is smooth of relative dimension one, proper, and geometrically irreducible.	Global challenge input; preserved by field base change.
H3	$\Omega = \text{SeparableClosure } K$, hence Ω/K is algebraic, separable, and normal and Ω is separably closed; every finite stage $E/K \subset \Omega$ admitted to R6 carries Module.Finite $_K E$ and Algebra.IsSeparable KE .	Used for translated rank-one coverage, finite-stage spreading, and the finite-Galois final stage. The finite normal closure is taken only after the separability certificate is present, and remains inside Ω . This is stronger than merely choosing an algebraic separably closed extension over an imperfect field; the exact Lean instantiation must expose the separability and normality needed at the normal-closure step.
H4	A legacy finite-stage glue package P_0 has final field $E_0 = P_0.N.1 \subset \Omega$.	Checked from finite presentation of the finite atlas under algebraicity alone; it supplies the comparison shape but neither a separability certificate nor the final route carrier.
H5	A legacy package P is selected jointly with the universal-class descent, with $L = P.N.1 \subset \Omega$ finite Galois over K .	The normal-closure theorem supplies the field enlargement; reflecting the whole finite diagram at that stage and producing the exact Lean package are planned parts of the first producer.
H6	$h_{J_L} \simeq \text{Pic}_{C_L}^0$ on the literal carrier $J_L = P.\text{gluedOver}$.	Unresolved first mathematical producer ; it must come from a class ξ_L on this carrier and its Yoneda equivalence.
H7	$P.\text{gluedMap}$ is projective.	Planned downstream geometry, blocked by H6 ; this is the one selected sufficient input for orbit-affineness.
H8	$J_L \rightarrow \text{Spec } L$ is locally of finite type and quasi-compact.	Produced from the finite affine presentation; the conditional Galois wrappers consume these certificates.
H9	A K -rational point $P \in C(K)$.	Used only by pointed Abel–Jacobi consumers in Phase 8; never used to construct Pic^0 .
H10	Properness, smoothness of relative dimension g , and geometric irreducibility of the final carrier.	Outputs still to be instantiated from the original-field datum; they are not hypotheses of representability.

Remark 2 (Status discipline). *Proved* means a public-root-reachable Lean declaration with the displayed type. *Imported* means a source or Mathlib theorem used with its full hypotheses. *Conditional* means Lean checks an implication whose antecedent remains visible; a `\leanok` marker on such a node certifies only that generic implication, not its unresolved route instance. *Planned* means the exact target type is fixed but no producer exists. *Unresolved* means that target is the first unmet mathematical obligation on the route. A source declaration not reachable from the public root is *unverified* here and receives no Lean marker.

Proof. This convention separates implementation state from mathematical truth and prevents a consumer theorem from being counted as its missing producer. $\square$

Chapter 2

Degree- g divisors and the rank-one Abel isomorphism

Throughout this chapter the base field is an arbitrary field k , and C/k satisfies H1–H2. No separable-closedness or rational point is used.

2.1 The represented divisor package

Theorem 3 (Represented widened divisors of degree g).

Let $\pi : C \rightarrow \mathbb{P}_k^1$ be the finite dominant map chosen internally from H1–H2. There is a k -scheme D_g and, naturally for every $T \rightarrow \operatorname{Spec} k$, a bijection

$$\operatorname{Hom}_k(T, D_g) \simeq \operatorname{divFunctorAff}(C, g)(T).$$

The right side is the Lean widened divisor functor: compatible divisor families on arbitrary affine opens, with the local certification used by the finite-map/cohomology construction. This statement must not be silently replaced by representability of a differently defined divisor functor.

Proof. Kleiman’s Theorem 3.7 represents relative effective divisors for a projective flat family as an open of the Hilbert scheme, and Exercise 3.8 isolates fixed degree on a curve. The Lean specialization chooses π , proves the required H^0 and H^1 finiteness and Euler-characteristic identities from H1–H2, constructs the finite-relation affine charts, and glues their classifiers. At parameter $g(C)$ this yields the displayed ‘RepresentableBy’ certificate on the exact functor ‘ $\operatorname{divFunctorAff} C$ (genus C)’. $\square$

Definition 4 (The admissible divisor source is auxiliary).

Lean also constructs a quasi-compact represented divisor source in a larger admissible degree. In this route it is used only to prove quasi-compactness of the final separably closed representer via a surjective Abel map. Its unrestricted Abel map is not the rank-one chart and is not quotiented to construct J .

Proof. The large degree kills H^1 and supplies etale-local effective representatives, exactly as in Kleiman’s bounded Abel construction. For degree greater than g , however, complete linear systems have positive dimension; hence the unrestricted Abel map cannot be an open immersion. $\square$

2.2 The intrinsic rank-one open

Theorem 5 (The rank-one Picard locus and its represented inverse image).

For a test T/k and $\lambda \in \text{Pic}_C^g(T)$, say that λ is rank one precisely when

$$\forall (A, [\text{CommRing } A], [\text{Algebra } k A]) \forall t : \text{Spec}_k A \rightarrow T, \quad \text{Nonempty}(\text{PicRankOneLocalPresentation}(\pi, \lambda_t)), \quad (\text{R1P})$$

where λ_t is the functorial pullback of λ . A witness in (R1P) is not an unnamed line bundle: it contains an étale cover $q : \text{Spec } B \rightarrow \text{Spec } A$, a descent-class representative u with

$$\text{PicEtAff.mk}(u) = \text{picEtAffineEquiv}(\lambda_t),$$

a `BasicOpenCocycleDatum` d satisfying $[u] = \text{relPicMk}(d.\text{cechPicClass})$, and a `Scheme.Modules` object M with an isomorphism from its underlying sheaf to $d.\text{sheaf}$. It also contains `Scheme.Modules.IsLineBundle`(M) and, for every cartesian square $X' \rightarrow X$ over $\text{Spec } B$,

$$\text{IsIso}((\text{canonicalBaseChangeMap } sq).\text{app } M),$$

together with $\text{Subsingleton}(H^1(d.\text{sheaf}))$, finite and projective $H^0(d.\text{sheaf})$ over B , and $\text{rankAtStalk}(H^0(d.\text{sheaf}), p) = 1$ for every $p \in \text{PrimeSpectrum } B$. These classes form a subfunctor

$$\text{Pic}_C^{g, \text{rk}=1} \hookrightarrow \text{Pic}_C^g$$

relatively represented by open immersions.

Pulling this subfunctor back along the degree- g Abel map $a_g : h_{D_g} \rightarrow \text{Pic}_C^g$ gives an actual open $W \subset D_g$, encoded by ‘`DivRankOneOpenData`’. For every T , a map $T \rightarrow D_g$ factors through W exactly when its Abel class is in the rank-one subfunctor.

Proof. The Lean proof does not assume an unidentified universal line bundle on the Picard functor. It defines the split locus of a class at residue-field points, proves that this locus pulls back exactly, proves openness on affine tests using a native étale presentation and flat quotient maps, and then glues over affine opens. The resulting relatively represented open is pulled back along the represented Abel transformation to obtain W . This realizes the open-locus specialization used in Kleiman’s proof while retaining the precise class-tied Lean predicate. $\square$

Theorem 6 (Canonical rank-one Abel isomorphism).

With $\pi = \text{divRepAffP1Map}(C)$, Lean’s primary statement is the isomorphism of sigma-extension objects

$$\text{rankOneDivisorLocus}(\pi) \xrightarrow{\sim} \text{rankOneLocus}(\pi).$$

The slicewise isomorphism `canonicalRankOneAbelSliceIso`, the represented divisor open $W = \text{divRankOneOpenOver}(\pi)$, and `picRankOneOpenSigmaIso` identify this with the restricted Abel transformation $a_g|_W : h_W \xrightarrow{\sim} \text{Pic}_C^{g, \text{rk}=1}$. Explicitly, for every test T/k and every rank-one class λ , the counit $f_T^* f_{T*} \mathcal{L} \rightarrow \mathcal{L}$ of any class-tied local presentation has a canonical zero divisor; these local divisors descend to a unique element of $W(T)$. Its Abel class is λ , and any two divisors in $W(T)$ with the same Abel class are equal.

Proof. Milne identifies an Abel fibre with the complete linear system and records that the generic degree- g locus has $h^0 = 1$. Lean strengthens the required family statement directly: rank-one local data provide the evaluation divisor on each affine; naturality and arbitrary-test uniqueness glue it; injectivity of the restricted Abel map turns its right inverse into a two-sided inverse. Sigma extension gives the displayed big-site isomorphism. Milne’s theorem is used only as the

classical mathematical model, not as a claim that the whole symmetric-power construction is the chosen formalization route. Milne Chapter III, Section 5 has standing assumptions that the curve is complete nonsingular, $g > 0$, and equipped with a rational point; none is silently transferred to the stronger familywise Lean statement or to the genus-zero case. $\square$

Chapter 3

The separably closed rank-one cover

Let k now be separably closed and retain H1–H2 for C/k . Separably closedness is used to construct translators and prove coverage; it is not inserted into the rank-one Abel isomorphism itself.

3.1 Translated charts

Definition 7 (The exact translator and chart family).

Put $\mathcal{P}_\Sigma^0 = (\text{pic0SigmaSheaf } C).1$. A translator is exactly an element of the Lean subtype

$$\text{PicRankOneTranslatorIndex}(C) = \{a : (C \otimes \text{overSpec}(k, k)).\text{left.CechPic} \mid \text{classDeg}_k(a) = g\}.$$

On the slice site, multiplication by its theta family gives the natural isomorphism

$$\tau_a(T) : \text{Pic}_C^0(T) \xrightarrow{\sim} \text{Pic}_C^g(T), \quad \lambda \mapsto \lambda \otimes a_T.$$

With W the represented rank-one divisor open from Theorem 5, the actual big-site chart is

$$j_a : \text{yoneda}(W.\text{left}) \longrightarrow \mathcal{P}_\Sigma^0.$$

It is the canonical rank-one Abel isomorphism, followed by the rank-one-locus inclusion, followed by the inverse of the sigma extension of τ_a . Every j_a is relatively represented by open immersions. The later representability theorem uses the canonical sigma-extension-to-slice bridge to recover the target $\text{pic0TypeFunctor}(C)$; the chart target itself is not definitionally that slice functor.

Proof. The three factors have the exact Lean carriers

$$\text{yoneda}(W.\text{left}) \xrightarrow{\sim} \text{rankOneLocus}(\pi) \longrightarrow \text{sigmaExtension}(\text{Pic}_C^g) \xrightarrow{\text{sigmaExtension}(\tau_a)^{-1}} \mathcal{P}_\Sigma^0.$$

The first and last factors are natural isomorphisms and the middle factor is the checked relatively represented open inclusion. Closure of open immersions under composition proves the claim. $\square$

Theorem 8 (Translated rank-one coverage).

The auxiliary producer has the following exact field-valued type. For every field extension K/k and every $\mu \in \text{pic0Subgroup}(C, \text{overSpec}(k, K))$, it returns a finite separable extension

L/K , the scalar-tower instances $k \rightarrow K \rightarrow L$, a finite-support datum on C_L , its residue-degree-one certificate, and the Čech subtraction equality; its final conclusion is that the translated layer element belongs to the rank-one open. In symbols, its witness has the shape

$$\begin{aligned} & \exists L [\text{Field } L] [\text{Algebra } kL] [\text{Algebra } KL] [\text{IsScalarTower } kKL] \\ & [\text{Module. Finite}_K L] [\text{Algebra. IsSeparable } KL] \exists d r, \\ & \text{classDeg}_k(\text{chartTwistClass}(d.m, r.Z)) = g \wedge \text{Finite}(r.S.\text{support}) \\ & \wedge (\forall p \in r.S.\text{support}, \text{residueDeg}_L(p) = 1) \\ & \wedge \text{picClass}_L(r.S) = \text{CechPic.map}(C, k, L)(\text{chartTwistClass}(0, r.Z)^{-1}) \\ & \wedge \text{rankOneLayer}(\mu, d) \in \text{PicRankOneOpen}(\pi)(\text{overSpec}(k, K)). \end{aligned}$$

The public theorem `exists_picRankOneTranslatedChart_fieldFactorization` deliberately hides that auxiliary L, d, r : its exact conclusion is only

$$\begin{aligned} & \exists a : \text{PicRankOneTranslatorIndex}(C) \exists x : (\text{overSpec}(k, K)).\text{left} \longrightarrow W.\text{left}, \\ & (j_a)_{(\text{overSpec}(k, K)).\text{left}}(x) = \langle (\text{overSpec}(k, K)).\text{hom}, \mu \rangle \\ & \in \mathcal{P}_{\Sigma}^0((\text{overSpec}(k, K)).\text{left}). \end{aligned}$$

Thus L is not a field-valued output of the public factorization and is never identified with k or K . Consequently, for every k -scheme T , every $s \in \mathcal{P}_{\Sigma}^0(T)$, and every point $t \in T$, there are an open neighbourhood $U \subseteq T$, an index a , and $x_U : U \rightarrow W.\text{left}$ with $j_a(x_U) = s|_U$. Equivalently, the family $\{j_a\}_a$ is pointwise covering and its sigma-desc map is Zariski-locally surjective.

Proof. Restrict the slice component of s to $\kappa(t)$. The translated-drop construction first produces the displayed finite separable extension $L/\kappa(t)$, a finite support on C_L of residue degree one over L , and the Čech subtraction equality. The resulting translated class is then proved to lie in the public `PicRankOneOpen`, and the canonical Abel inverse gives the factorization through W . There is no step in which separable closedness identifies the auxiliary residue field with k , and no descent of the witness from L (or from $\kappa(t)$) to k is claimed. Relative representability of j_a pulls this residue-field factorization back to an open neighbourhood of t . Stacks Tag 056U is used only for density of closed points with finite separable residue field on a smooth scheme; obtaining the required rational point over a separably closed field is a separate specialization.

Milne’s Lemma 5.2(b) is the classical fibrewise model: a general degree- g divisor has $h^0 = 1$. The Lean proof is stronger in the direction needed here because it constructs the translator and proves membership in the exact étale-local rank-one predicate. Milne’s surrounding Section III.5 assumes a complete nonsingular curve of positive genus and a chosen rational point; the citation is only the fibrewise model, while the checked translated-drop theorem retains AJCR’s displayed hypotheses and introduces no such point. $\square$

3.2 The exact separably closed representer

Theorem 9 (Representability from the translated atlas).

There is one selected pair

$$(J_k, r_k), \quad J_k \longrightarrow \text{Spec } k, \quad r_k : h_{J_k} \xrightarrow{\sim} \text{Pic}_C^0,$$

where J_k is obtained by gluing the represented open charts $\{W \xrightarrow{j_a} \text{Pic}_C^0\}_a$. In Lean this pair is the dependent value

$$\text{pic0_sepClosed_representableBy}(C) : \sum_{J : \text{Over}(\text{Spec } k)} \text{RepresentableBy}(\text{pic0TypeFunctor}(C), J).$$

Proof. Stacks Tag 01JJ says that a Zariski sheaf covered by representable subfunctors whose inclusions are relatively represented by open immersions is representable. Apply it to the charts of Definition 7: openness is proved there and coverage is Theorem 8. The overlap maps are forced by equality of their universal Picard classes, so the glued functor of points is naturally Pic_C^0 . Kleiman’s Theorem 4.8 supplies the surrounding classical Picard representability theorem, but the checked Lean result is specifically the translated rank-one atlas specialization just described. $\square$

Theorem 10 (Finiteness and same-carrier data over a separably closed field).

The selected structure morphism $J_k \rightarrow \text{Spec } k$ is locally of finite presentation and quasi-compact, and J_k is quasi-separated. The exact same pair (J_k, r_k) defines

$$\text{PicRepDatum}(k, k, C) \quad \text{and} \quad \text{JacobianData}(C);$$

the second adds the quasi-compactness certificate, while neither handoff changes the carrier, representation, group object, or universal element.

Proof. Each rank-one chart is locally of finite type, so the glued structure map is locally of finite type. Since $\text{Spec } k$ is Noetherian, Stacks Tag 01TX upgrades this to local finite presentation. The represented admissible divisor source is quasi-compact and its Abel morphism to this exact J_k is étale-locally surjective, hence surjective on underlying points; quasi-compactness descends through that map. Local finite type over a field makes J_k locally Noetherian and therefore quasi-separated in the checked interface. Finally the definitional projection lemmas for `picRepDatumSepClosed` and `jacobianDataSepClosed` identify both fields with (J_k, r_k) literally. $\square$

Chapter 4

Finite-stage carrier and base-change comparison

Return to the original curve C/K . Fix the separable closure Ω/K of Hypothesis H3, put $C_\Omega = C \times_K \Omega$, and apply Chapter 3 to obtain the exact pair (J_Ω, r_Ω) . The finite-stage package is parameterized in the Lean order `Pic0FiniteStageGluePackage` C_Ω K : its first curve argument is over the separably closed field Ω , while its ambient field parameter is the original K , with `[Algebra  $K$   $\Omega$ ]` and `[Algebra.IsAlgebraic  $K$   $\Omega$ ]`. Thus every displayed $P_0.N.1$ is a finite subextension of Ω/K , not a finite extension of an unspecified field.

4.1 A finite affine atlas

Theorem 11 (Finite-presentation atlas and simultaneous models).

The atlas structure first chooses a finite set I of affine opens $U_i \subset J_\Omega$ covering J_Ω . It records compactness of every $U_i \cap U_j$, a finite affine overlap presentation of that intersection, and finite presentation of each chart ring $A_i = \Gamma(U_i, \mathcal{O})$. Its first simultaneous-model theorem models only this chart-ring family over one finite subextension.

Separately, separatedness of the represented Picard group makes the exact ordered intersection $U_{ij} := U_i \cap U_j$ affine. Define

$$A_i = \Gamma(U_i, \mathcal{O}), \quad B_{ij} = \Gamma(U_{ij}, \mathcal{O}).$$

The package index is exactly

$$\text{Pic0FiniteStageRingIndex} = I \sqcup (I \times I),$$

with ring A_i on the left summand and B_{ij} on the right. Every ring in this combined family is finitely presented over Ω , and one finite subextension models all of them simultaneously. The two canonical algebra maps $A_i \rightarrow B_{ij}$ and $A_j \rightarrow B_{ij}$ are defined and finitely indexed, but their finite-stage models, transitions, and equations are not conclusions of either ring-model theorem; those belong to the `GluePackage` below.

Proof. Quasi-compactness gives the finite affine cover, and separatedness of the represented group makes each ordered intersection affine (hence quasi-compact). Local finite presentation gives finite presentation of the selected chart and exact-overlap coordinate rings. Apply `Stacks Tag 01ZM` to the inverse system of affine field stages: its transition maps are affine, and the schemes

involved are quasi-compact and quasi-separated. The first attached producer descends only the chart rings; the second applies the same finite-family theorem to ‘Pic0FiniteStageRingIndex’ and descends all chart and exact pair-overlap rings. Restriction maps, inverse transitions, and their equations are the separate next package, not outputs of either attachment. $\square$

Theorem 12 (The finite-stage GluePackage).

Under only [Algebra.IsAlgebraic $K \Omega$], there is a package P_0 with nested finite subextensions

$$K \subseteq P_0.L.1 \subseteq P_0.M.1 \subseteq E_0 := P_0.N.1 \subseteq \Omega.$$

It contains finite-presentation models of every A_i and canonical B_{ij} , all algebra maps in the restriction and reversed-overlap transition families, proofs that restriction maps remain open immersions, and the descended triple-overlap equations. The package does not contain or produce an Algebra.IsSeparable $K E_0$ or Galois certificate. These data form

$$P_0.\text{glueData} : \text{Scheme.GlueData}$$

and a glued structure morphism

$$P_0.\text{gluedMap} : P_0.\text{glueData}.glued \longrightarrow \text{Spec } E_0.$$

Set $J_{E_0} = P_0.\text{gluedOver} := \text{Over.mk}(P_0.\text{gluedMap})$. Its structure map is locally of finite type and quasi-compact.

Proof. Use Theorem 11 for the finite list of chart and exact-overlap rings. The ‘Pic0FiniteStageRestrictionIndex’ supplies the two restriction legs for every ordered pair, while the transition and triple indices supply the reversal and cocycle fields. Enlarge to $P_0.M.1$ so every restriction and transition map descends and every required open-immersion property holds; Stacks Tag 0EUU is the finite-stage open-immersion principle, under its locally-finitely-presented stage-map hypothesis. Enlarge to $P_0.N.1$ so inverse and triple-cocycle equations hold; Tag 081E is the corresponding isomorphism descent principle, again with its finite-presentation hypothesis. The contravariant spectra of the descended rings and maps satisfy the exact Scheme.GlueData fields, whose standard gluing construction gives J_{E_0} . Its finite affine cover proves quasi-compactness, and finite type of the descended chart algebras proves local finite type of its structure map. $\square$

4.2 Nested pullbacks and overlap comparison

Theorem 13 (Finite-stage overlap diagrams recover the original atlas).

For every pair i, j , let G_{ij} be the (i, j) -object of the base-changed finite-stage GlueData, and let V_{ij} be the (i, j) -object of the canonical pic0SepClosedAtlasGlueData(C_Ω). The exact checked comparison is

$$\gamma_{ij} : G_{ij} \xrightarrow{\sim} V_{ij}.$$

It is the composite of the associativity/flattening isomorphism for the nested tensor products, the comparison with the selected finite affine overlap cover, and the pullback isomorphism induced by the descended restriction maps, ending with

$$\delta_{ij} : U_{ij}.toScheme \xrightarrow{\sim} V_{ij}.$$

Thus $\delta_{ij}^{-1}\gamma_{ij}$, not γ_{ij} alone, identifies the base-changed object with the exact intersection $U_i \cap U_j$. Under γ_{ij} , both projections agree exactly with the left and right atlas restriction maps. The transition maps agree on triple overlaps.

Proof. On coordinate rings, associativity identifies $\Omega \otimes_{E_0} (E_0 \otimes_{P_0.M.1} (P_0.M.1 \otimes_{P_0.L.1} A))$ with the original Ω -algebra. The package comparison equations identify both restriction homomorphisms and the cyclic transition homomorphism after this flattening. Passing contravariantly to spectra gives the nested-pullback comparison and then the isomorphism to the canonical Glue-Data object. The separate ‘pic0SepClosedAtlasOverlapIso’ identifies the exact intersection with that canonical object. The named first- and second-projection theorems, followed by the triple face equation, verify the complete gluing diagram; no appeal to Picard representability occurs in this argument. $\square$

Theorem 14 (Global finite-stage base-change comparison).

The chart comparisons glue to an isomorphism over Ω

$$\beta_{P_0} : J_{E_0} \times_{\mathrm{Spec} E_0} \mathrm{Spec} \Omega \xrightarrow{\sim} J_\Omega.$$

It commutes with the two structure maps to $\mathrm{Spec} \Omega$. The attached proved assertion is precisely this carrier isomorphism and its structure-map equation. Transporting r_Ω along β_{P_0} consequently gives

$$\mathrm{RepresentableBy}(\mathrm{Pic}_{C_\Omega}^0, J_{E_0} \times_{E_0} \Omega).$$

*This formal consequence is not credited as a finite-stage representation by the attachment. In particular it does **not** prove $\mathrm{RepresentableBy}(\mathrm{Pic}_{C_{E_0}}^0, J_{E_0})$.*

Proof. Theorem 13 identifies the complete base-changed gluing diagram with the finite atlas of J_Ω . Uniqueness of gluing yields the first global isomorphism; the checked structure-map equation upgrades it to the slice category. The last displayed post-base-change certificate is ordinary transport of an existing representer across that slice isomorphism. Since no Picard class on J_{E_0} and no natural equivalence over E_0 has yet been constructed, interpreting this result as finite-stage representability would be a logical error.

The source tree also contains the wrappers `finiteStageBaseChangeOverIso` and `pic0RepresentableBy_finiteS`. Their defining module is not reachable from the public `AlgebraicJacobian` root, so they are unverified implementation artifacts here and receive no `\leanok` credit. Their targets in any case remain the slice comparison and post-base-change representation over Ω . $\square$

Chapter 5

Universal class and the finite-stage representability gap

Chapter 4 proves the carrier comparison for any legacy $P_0 : \text{Pic0FiniteStageGluePackage}(C_\Omega, K)$. The route does not freeze that arbitrary stage. Instead, the first missing producer must choose a final legacy package only after all carrier, class, and cocycle coefficients have been enlarged to one finite Galois subextension. This joint choice is what prevents a silent change of carrier.

5.1 What is pinned over the separably closed field

Theorem 15 (The universal class and all atlas restrictions over Ω).

Let

$$\xi_\Omega := r_\Omega(\text{id}_{J_\Omega}) \in \text{Pic}_{C_\Omega}^0(J_\Omega).$$

For every selected affine chart $U_i \subset J_\Omega$ and every exact ordered pair-overlap $U_{ij} = U_i \cap U_j$, the checked classes are exactly the pullbacks of this one ξ_Ω . If $q_{ij}^i : U_{ij} \rightarrow U_i$ and $q_{ij}^j : U_{ij} \rightarrow U_j$, then

$$(q_{ij}^i)^*(\xi_\Omega|_{U_i}) = \xi_\Omega|_{U_{ij}} = (q_{ij}^j)^*(\xi_\Omega|_{U_j}).$$

Proof. Apply the representing equivalence to the identity of J_Ω . Its naturality says that precomposition by a chart or overlap morphism is pullback of the resulting Picard class, proving both restriction equations. Every class in this theorem still lives over Ω . $\square$

The source tree additionally declares `pic0ClassYoneda` and `pic0SepClosedClassYoneda_app`, which express the expected class-to-natural-transformation formula. Their defining module is not reachable from the public `AlgebraicJacobian` root and has no current verified artifact. They are therefore unverified implementation aids, not attachments or proof credit in this specification.

5.2 The first unmet producer

Theorem 16 (Joint Galois-stage package and descended universal class).

There must exist, simultaneously:

1. a legacy package $P : \text{Pic0FiniteStageGluePackage}(C_\Omega, K)$;

2. for $L := P.N.1 \subset \Omega$, certifies that L/K is finite Galois, with the displayed inclusion and scalar-tower maps;
3. on the literal carrier $J_L := P.\text{gluedOver}$, a class

$$\xi_L \in \text{Pic}_{C_L}^0(J_L) = \text{pic0Subgroup}((\text{baseChange}_{K,L})C, P.\text{gluedOver}), \quad C_L = C \times_K L.$$

The symbol P in this theorem is a new, jointly selected package; it is not the arbitrary comparison witness P_0 from Theorem 14. If the construction starts from P_0 , it must additionally provide an explicit refinement map from P_0 to P together with the induced Over-map and its compatibility with every chart and overlap comparison. Without those fields, P_0 is discarded after the comparison theorem and no field enlargement is allowed to silently replace its carrier. All subsequent occurrences of J_L , ξ_L , (FS-Rep), (Proj), and (OA) quantify over this single final P . For the checked comparison $\beta_P : J_L \times_L \Omega \xrightarrow{\sim} J_\Omega$, this class must satisfy

$$\text{bc}_{L,\Omega}(\xi_L) = \beta_P^*(\xi_\Omega) \quad \text{in} \quad \text{Pic}_{C_\Omega}^0(J_L \times_L \Omega). \quad (\text{UC})$$

Equivalently, its restriction to every descended chart pulls back to $\xi_\Omega|_{U_i}$, and its two restrictions to every descended overlap agree before gluing.

Proof. The field-theoretic part has no route choice. Every finite coefficient set lies in a finite separable subextension because Ω/K is algebraic and separable. Normality of Ω/K and Stacks Tag 0BMG place a finite normal enlargement L/K , containing that coefficient field, inside Ω . Separability inherited from Ω/K , equivalently Tag 0EXM applied to the normal closure of the finite separable input, makes L/K finite Galois. Only then re-run the finite-presentation descent of all chart rings, restriction maps, inverse maps, and triple equations at L , and assemble the final legacy package with $P.N.1 = L$. One must not retain an earlier package while changing its field.

The unresolved mathematical step is to descend finite presentations of representatives of the finitely many chart classes, all transition isomorphisms and their cocycle equations at that same L , glue them using the actual etale-sheafified relative Picard equivalence relation, and prove degree zero after every field-valued pullback. Comparison through β_P must then prove (UC). Tag 01ZM supplies finite-presentation, morphism, and equality descent. Section 44.4 (Tag 0B9K) describes the fppf Picard sheaf and invertible-sheaf descent, but is only orientation: it does not identify that object with AJCR's big-etale degree-zero carrier or construct ξ_L . The planned Lean handoff is one dependent package containing exactly P , the finite-Galois certificates, ξ_L , and equation (UC). $\square$

Theorem 17 (Yoneda equivalence on the same finite Galois carrier).

For the same (P, L, ξ_L) , every L -scheme T must admit the bijection

$$\Phi_T : \text{Hom}_L(T, J_L) \xrightarrow{\sim} \text{Pic}_{C_L}^0(T), \quad f \mapsto f^*\xi_L. \quad (\text{Y})$$

For every $q : T' \rightarrow T$ and $f : T \rightarrow J_L$, it must satisfy the correctly typed naturality equation

$$\Phi_{T'}(f \circ q) = q^*\Phi_T(f).$$

The exact public Lean handoff is

$$\text{RepresentableBy}(\text{pic0TypeFunctor}(C_L), P.\text{gluedOver}). \quad (\text{FS-Rep})$$

Proof. Pullback of ξ_L defines the transformation in (Y). To prove every component bijective, compare it after the faithfully flat extension $L \rightarrow \Omega$. Equation (UC), β_P , and r_Ω identify the result with a bijection. Existence and equality of morphisms then descend; Stacks Remark 35.35.12 (Tag 040L) supplies the abstract fpqc morphism-descent principle. The specialization must be performed uniformly for arbitrary T , including the curve scalar-tower and big-etale degree-zero transports.

No public-root declaration proves (Y) or (FS-Rep). In particular, `pic0RepresentableBy_finiteStageBaseChange` has target over Ω , not over L , and its own module is unrooted. $\square$

Remark 18 (Frontier accounting). The first unmet mathematical producer is the class ξ_L and equation (UC) in Theorem 16. Choosing the finite normal coefficient field and rebuilding the finite diagram there is a fixed imported/planned prelude inside that same package, not a competing route. The frontier moves only when a class on the displayed literal carrier exists; further transport or comparison wrappers do not move it.

Chapter 6

Finite-Galois descent to the original field

Fix the single witness (P, L, ξ_L) from Theorem 16, where $L = P.N.1$ is finite Galois over K , and assume Theorem 17 has supplied

$$r_L : h_{J_L} \xrightarrow{\sim} \text{Pic}_{C_L}^0, \quad J_L = P.\text{gluedOver}.$$

No object is base-changed or replaced in this chapter: the Galois action, projectivity input, orbit condition, and quotient all use this literal carrier.

6.1 The one selected exact-carrier geometry route

The equivalence r_L canonically equips J_L with a commutative group object and with the semi-linear action

$$\rho : \text{Gal}(L/K) \longrightarrow \text{Aut}_{K\text{-semilin}}(J_L)$$

characterized by invariance of the universal Picard class.

Theorem 19 (Projectivity of the literal finite-stage carrier).

Construct the certificate

$$h_{\text{proj}} : \text{IsProjective}(P.\text{gluedMap}), \quad (\text{Proj})$$

where P is the same legacy package that carries ξ_L and r_L . This is the sole selected producer for the orbit condition. A proof by an independently chosen projective-space immersion is not an alternate route.

Proof. This target is planned and blocked on (FS-Rep). A valid proof must construct a relatively ample invertible sheaf, or equivalently the finite-type closed immersion furnished by the projectivity API, on the literal glued scheme and verify compatibility with $P.\text{gluedMap}$. Classical projectivity of the Jacobian cannot be imported here until its hypotheses and its identification with AJCR's degree-zero big-etale functor are proved. In particular, projectivity of an admissible divisor source, a stable-facade carrier, or an object merely isomorphic after base change does not inhabit (Proj). $\square$

Theorem 20 (Projectivity implies exact-carrier orbit-affineness).

From (Proj) , for every scheme point $x \in |J_L|$, produce an affine open $V \subseteq J_L$ such that

$$\{\rho(\sigma)(x) : \sigma \in \text{Gal}(L/K)\} \subseteq V. \quad (\text{OA})$$

The exact Lean output is

`(pic0SemilinearGalActionOfRepresentableBy(C, r_L)).OrbitsInAffineOpen.`

Proof. The finite Galois orbit is a finite subset. Projectivity supplies a closed immersion of J_L into finite relative projective space; choose a hyperplane avoiding the orbit, and pull back its affine complement. The source tree contains the exact conditional wrapper `pic0FiniteStageOrbitsInAffineOpen_of_isP`
`C Ck P repl hproj`. It has the required binders `[FiniteDimensional K P.N.1] [IsGalois K P.N.1]` and returns (OA) on `P.gluedOver`. Its module is not public-root reachable, so it receives no `\leanok` marker; rooting that wrapper is a planned integration step after the planned input (Proj) . The immersion entry point used internally by the wrapper is quarantined as a second public route. $\square$

6.2 The conditional descent theorem

Theorem 21 (Finite-Galois descent of the represented degree-zero functor).

Let L/K be the displayed finite Galois extension, with the Lean instances `[Algebra K L]`, `[FiniteDimensional K L]`, and `[IsGalois K L]`. Let $C_L = (\text{baseChange } K L).obj C$, let $J_L : \text{Over}(\text{Spec } L)$, and assume the exact input

$$r_L : \text{RepresentableBy}(\text{pic0TypeFunctor}(C_L), J_L)$$

together with the typeclass `OrbitsInAffineOpen` for `pic0SemilinearGalActionOfRepresentableBy(C, r_L)`. The stable affine opens glue to

$$J := \text{StableAffineOpen}.gluedQuotientOver(\rho) \longrightarrow \text{Spec } K.$$

For every K -scheme T , the checked equivalences are

$$\text{Hom}_K(T, J) \simeq \text{Hom}_L(T_L, J_L)^{\text{Gal}(L/K)} \simeq \text{Pic}_{C_L}^0(T_L)^{\text{Gal}(L/K)} \simeq \text{Pic}_C^0(T). \quad (\text{G})$$

Hence J represents `pic0TypeFunctor(C)`. If $J_L \rightarrow \text{Spec } L$ is locally of finite type, respectively quasi-compact, then so is $J \rightarrow \text{Spec } K$.

Proof. Orbit-affineness chooses a Galois-stable affine cover. Affine fpqc descent glues its invariant coordinate rings; Stacks Section 35.38 (Tag 0246) records effective descent for quasi-affine morphisms and Tag 040L records descent of compatible morphisms. The first equivalence in (G) is the checked quotient universal property. The second uses r_L and its canonical semilinear action. The third is the checked equivalence between original-field Picard classes and Galois-invariant classes after base change. Their naturality in T is exactly the `RepresentableBy` certificate, and the finiteness properties descend through the same quotient.

This is a checked *conditional* theorem: the displayed r_L , field typeclasses, and `OrbitsInAffineOpen` remain visible inputs. Its Lean binder for the invariant comparison includes geometric reducedness of C ; smoothness of the challenge curve derives that instance, so H2 already discharges it. $\square$

Remark 22 (No carrier switch). The selected route finalizes one legacy package with $L = P.N.1$ before descending ξ_L . It then uses $P.\text{gluedOver}$ in (FS-Rep), (Proj), (OA), and the generic finite-Galois theorem. The newer `Pic0FiniteStageStableGluePackage` has no checked carrier comparison with this object and is never substituted. Forward representer transport along an arbitrary object isomorphism is likewise a consumer of an existing representation, not a producer for R6.

Chapter 7

Same-carrier data and Phase 8 consumers

7.1 The public data boundary

Definition 23 (Exact PicRepDatum and JacobianData carriers). For fields $k \rightarrow k'$ and a curve C'/k' , a $\text{PicRepDatum}(k, k', C')$ consists exactly of

$$J'/k', \quad r' : h_{J'} \xrightarrow{\sim} \text{Pic}_{C'}^0, \quad J' \rightarrow \text{Spec } k' \text{ locally of finite type.}$$

For a curve C/k , a $\text{JacobianData}(C)$ consists exactly of the same-field carrier and representation, the same local-finite-type certificate, and a quasi-compactness certificate for its structure map. In both structures the group object, the bijections $\text{Hom}(T, J') \simeq \text{Pic}_{C'}^0(T)$, and uniqueness of the representing object are derived from r' ; they are not new fields.

Proof. This is a direct expansion of the two Lean structures. In particular, neither stores properness, smoothness, a dimension, an Abel source, or an Albanese theorem. Those are downstream consumers with separate inputs. $\square$

Theorem 24 (Conditional original-field package and literal handoff).

Under the exact hypotheses of Theorem 21, let J/K be its glued quotient. The locally-finite-type conclusion packages (J, r) as $d : \text{PicRepDatum}(K, K, C)$. Given the descended quasi-compactness certificate, define

$$D = d.\text{toJacobianData} : \text{JacobianData}(C).$$

Then, definitionally,

$$D.J = d.J = J, \quad D.\text{rep} = d.\text{rep} = r, \quad D.\text{grpObj} = d.\text{grpObj}, \quad D.\text{homEquiv} = d.\text{homEquiv}.$$

Proof. The checked finite-Galois wrapper builds the quotient and its representation, and descends local finite type and quasi-compactness from J_L . The ‘toJacobianData’ constructor assigns the four displayed fields without transport; all four comparison lemmas are proved by reflexivity. This is the required same-carrier handoff. It remains conditional until the first unresolved producer in Chapter 5 and the downstream projectivity/orbit producer in Chapter 6 are supplied. $\square$

7.2 Checked consumers of an arbitrary datum

Theorem 25 (Universal-element, Abel, and functorial consumers).

For any $D : \text{JacobianData}(C)$, the representation gives a commutative group object on $D.J$. For every K -point P of C , the class of $\mathcal{O}(\Delta - P \times C)$ defines

$$\text{ofCurve}_D(P) : C \longrightarrow D.J, \quad \text{ofCurve}_D(P) \circ P = e_{D.J}.$$

A morphism of curves $X \rightarrow Y$, together with data D_X, D_Y , induces the contravariant group morphism $D_Y.J \rightarrow D_X.J$, with checked identity and composition laws. For every field extension $K \rightarrow L$, $D.\text{baseChange}(L)$ has literal carrier $D.J \times_K L$, and for any datum D_L for C_L , uniqueness of representers gives the group-scheme isomorphism

$$\text{baseChangeIsoOfData}(D, D_L) : D.J \times_K L \xrightarrow{\sim} D_L.J.$$

Proof. All assertions are formal consequences of the represented degree-zero functor. The Abel map is the inverse representing equivalence applied to the diagonal class; naturality proves the pointing equation. Pullback of Picard classes supplies the contravariant curve map. Field base change uses the Picard base-change equivalence, and the displayed isomorphism is uniqueness of representing objects upgraded to group objects. Milne's Proposition 6.1 explains the intended pointed Jacobian map, but its Albanese factorization is not asserted in this theorem. $\square$

Theorem 26 (Tangent-space interface at the identity).

Let $e : \text{Spec } K \rightarrow D.J$ be the group identity, let $R = \mathcal{O}_{D.J,e}$, $\kappa(e) = R/\mathfrak{m}_e$, and $V = (\mathfrak{m}_e/\mathfrak{m}_e^2)^\vee$. There are checked bijections

$$V \simeq T_e D.J \simeq \ker(\text{Pic}_C^0(\text{Spec } K[\epsilon]) \longrightarrow \text{Pic}_C^0(\text{Spec } K)),$$

These are bijections of sets. In particular, the displayed kernel equivalence alone does not identify scalar multiplication or prove a dimension formula. Finite-dimensionality of the cotangent space is used internally by the checked numerical consumers below; it is not part of this theorem-family claim.

Theorem 27 (Conditional tangent dimension consumers).

If

$$\dim_{\kappa(e)}(\mathfrak{m}_e/\mathfrak{m}_e^2) = g(C), \tag{TD}$$

then $\dim_{\kappa(e)} V = g(C)$. If in addition $H^1(C, \mathcal{O}_C)$ is finite-dimensional over K , there exists an additive equivalence

$$\mathfrak{m}_e/\mathfrak{m}_e^2 \simeq_+ H^1(C, \mathcal{O}_C).$$

Both checked theorems take (TD) as an explicit hypothesis.

Theorem 28 (Semilinear epsilon-kernel comparison).

Construct a bijection $i : \kappa(e) \rightarrow K$, an additive equivalence

$$j : V \xrightarrow{\sim_+} H^1(C, \mathcal{O}_C),$$

and prove, for every $a \in \kappa(e)$ and $v \in V$,

$$j(av) = i(a)j(v). \tag{SL}$$

Then the checked scalar-bijection finrank lemma gives (TD), which feeds the two conditional consumers above.

Proof. Proof obligation. Kleiman Theorem 5.11 identifies the tangent space of the Picard scheme with $H^1(C, \mathcal{O}_C)$ by analysing line bundles over the dual numbers. The AJCR specialization must perform that calculation inside the epsilon-kernel of the exact etale-sheafified degree-zero functor, prove additivity, and identify the scalar action through the rational identity section. The present Lean equivalence is set-valued, so it cannot supply (SL) or (TD). $\square$

This tangent branch is the selected numerical input for the Phase 8 dimension producer below. It is not an alternate construction of the Jacobian: it starts from the same original-field datum and supplies only the missing relative dimension count.

Theorem 29 (Conditional Abel map under field base change).

For a field extension $K \rightarrow L$, a point $P : \text{Spec } K \rightarrow C$, and data D, D_L for C, C_L , assume `abelCrossBaseCechCore` $K L C P$: transport of the Cech class $[\mathcal{O}(\Delta)] - [\mathcal{O}(P \times C)]$ along the pullback counit and the scalar-tower comparison equals the corresponding class on C_L . Then

$$\text{baseChangeIsoOfData}(D, D_L) \circ (\text{ofCurve}_D(P))_L = \text{ofCurve}_{D_L}(P_L). \quad (\text{AJ-BC})$$

Equality of the transported diagonal-graph and constant-graph classes is a sufficient checked hypothesis for the Cech core.

Theorem 30 (The Abel Cech base-change core).

Prove `abelCrossBaseCechCore` $K L C P$ for every field extension $K \rightarrow L$, every challenge curve C/K , and every K -point P . This is the first unmet producer for (AJ-BC). The graph-factor theorem is a reduction to two equalities, not a proof of either equality.

Proof. Proof obligation. Compare the local equations of the diagonal and constant graph before and after the nested curve/test base change, prove their Cech cocycles correspond, then use multiplicativity and inversion to obtain the Abel class equation. Only after this theorem may the conditional datum-level equality be used in the challenge assembly. $\square$

Theorem 31 (Coherent base change for the selected datum family).

Select the datum D_C from the same R8–R9 construction for every challenge curve and every field extension. For $K \rightarrow L$, define

$$\gamma_{C,L} := \text{baseChangeIsoOfData}(D_C, D_{C_L}) : D_C \cdot J \times_K L \xrightarrow{\sim} D_{C_L} \cdot J.$$

Prove that $\gamma_{C,K}$ is the canonical identity comparison and, for $K \rightarrow L \rightarrow M$, that $\gamma_{C,M}$ equals the composite of the canonical pullback associator with $(\gamma_{C,L})_M$ and $\gamma_{C_L,M}$. Prove these as group-scheme equalities and prove (AJ-BC) for every $P \in C(K)$. These are exactly the producers for `Jacobian.baseChangeIso`, `baseChangeIso_id`, `baseChangeIso_comp`, and `baseChange_ofCurve`.

Proof. The object isomorphism exists by uniqueness of representers, but no current datum theorem supplies its identity and tower equations. Normalize all three isomorphisms by the universal Picard class and use extensionality of representing morphisms to prove the two coherences. The Abel equation is the checked conditional theorem instantiated by the Cech producer from Theorem 30. All equations must use the same selected data; arbitrary isomorphic replacement is not allowed. $\square$

Theorem 32 (Exact producers for the Phase 8 geometry interface).

For the selected original-field datum D_C , construct these three exact inputs on its unchanged carrier.

1. Put $n_C = \text{divRepAffAdmissibleParameter}(C)$. Construct the proper degree- n_C effective-divisor scheme D_C^{prop} , its geometric-irreducibility certificate, and the translated Abel morphism

$$a_C : D_C^{\text{prop}} \longrightarrow D_C.J$$

obtained from the universal divisor class using the fixed `admissibleAbelChartIndex/admissibleAbelChartDiv` translation. Prove a_C surjective and package precisely

$$A_C : \text{AbelSourceData}(D_C)$$

with fields $(D_C^{\text{prop}}, \text{IsProper}, \text{GeometricallyIrreducible}, a_C, \text{Surjective}(a_C))$.

2. Construct $h_{\text{red}} : \text{GeometricallyReduced}(D_C.J \rightarrow \text{Spec } K)$.
3. Construct a Zariski zero-hypercover $\mathcal{U}_C$ of $D_C.J$ and, for every leg i , a certificate

$$\text{SmoothOfRelativeDimension}(g(C), \mathcal{U}_C.f(i) \gg D_C.J.\text{hom}). \quad (\text{DimCov})$$

Proof. These are planned downstream producers, not assumptions to erase. For the first, Kleiman’s divisor/Hilbert construction is the classical source shape; the AJCR proof must build the actual proper divisor scheme rather than reuse the widened affine-test representer, define the displayed fixed translation, and specialize the Riemann inequality to prove surjectivity on scheme points. This downstream source is not an alternate construction of $D_C.J$.

For the second, specialize Kleiman Lemma 5.1 to the exact degree-zero functor: prove reducedness at the identity from the dual-number computation and move it by the checked translations, including arbitrary residue-field points. For the third, use (SL) to fix the identity-chart dimension at $g(C)$, use geometric reducedness and the group-scheme smoothness criterion to obtain a standard-smooth identity chart, and construct the translated Zariski cover with (DimCov). The Lean handoff is exactly the three arguments consumed below; no theorem currently constructs them. $\square$

Theorem 33 (Conditional geometric consumers).

Every datum carrier is separated. If an $\text{AbelSourceData}(D)$ is supplied—a proper geometrically irreducible K -scheme mapping surjectively to $D.J$ —then $D.J$ is proper and geometrically irreducible. If $D.J$ is geometrically reduced, then it is smooth. A Zariski zero-hypercover whose legs are smooth of relative dimension $g(C)$ proves

$$\text{SmoothOfRelativeDimension}(g(C), D.J \rightarrow \text{Spec } K).$$

The name `smoothOfRelativeDimension_statement` is only an abbreviation for this proposition, not a proof or a producer.

Proof. The unit of a group scheme over a field is closed, so the diagonal is closed and the carrier is separated. Properness and geometric irreducibility descend along the surjective Abel source in the checked consumer theorems. A geometrically reduced group scheme locally of finite type over a field is smooth. Relative dimension is local on the source, so it glues from the displayed hypercover. Each implication is checked, but none constructs its own Abel source, geometric-reducedness proof, or dimension chart. $\square$

7.3 The blocked Phase 8 assembly

Theorem 34 (Rational-extension and uniqueness substrate).

Over an algebraically closed field, a rational map from a smooth integral variety to an abelian variety extends uniquely once the displayed pure-codimension-one indeterminacy input is supplied; the checked Milne–3.3 theorem supplies that input under its stated variety and group hypotheses. Separately, if

$$\exists u : D.J \rightarrow A, \quad f = u \circ \text{ofCurve}_D(P) \quad (\text{Hex})$$

and uniqueness of the corresponding morphism is known after base change, the checked faithful-base-change theorem upgrades (Hex) to the required $\exists!$ over K . It also accepts an independently constructed datum over the extension, using uniqueness of representers to compare carriers.

Proof. Milne Theorem I.3.2 combines codimension at least two for maps into proper targets with Lemma I.3.3, which says that nonempty indeterminacy into a group variety is pure of codimension one; hence the locus is empty and the rational map is regular. For the descent statement, flat surjective base change is faithful on morphisms, so two factors equal after base change are equal over K . Neither argument constructs the factor in (Hex); that is a separate positive-genus obligation. $\square$

Theorem 35 (Conditional genus-zero package).

If

$$\forall T/K, \quad \text{Pic}_C^0(T) = \{0\}, \quad (\text{V0})$$

the checked constructor represents this functor by $\text{Spec } K$, and any datum representing it has terminal carrier. If $C \rightarrow \text{Spec } K$ is surjective, $P \in C(K)$, and the explicit existence statement (Hex) is supplied, then the genus-zero factor is unique.

Proof. Under (V0), the Yoneda functor of the terminal object and the degree-zero Picard functor are both singleton-valued, giving the datum and terminality. The Abel map to a terminal carrier is an epimorphism for a nonempty curve, so precomposition proves uniqueness. Existence is not formal: it requires the map $C \rightarrow A$ to be constant. Milne I.3, Proposition 3.9 proves this only for rational maps from $\mathbb{A}^1$ or $\mathbb{P}^1$. Its specialization here therefore first requires the planned pointed genus-zero classification $C \simeq \mathbb{P}_K^1$, or an equivalent rational-parametrization bridge. No checked theorem currently proves that bridge, derives (V0) from $g(C) = 0$, or supplies (Hex); all three remain explicit inputs. $\square$

Theorem 36 (The pointed Albanese consumer).

For a K -point $P : \text{Spec } K \rightarrow C$, an abelian variety A/K , and $f : C \rightarrow A$ with $f \circ P = e_A$, prove

$$\exists! u : D.J \rightarrow A, \quad f = u \circ \text{ofCurve}_D(P). \quad (\text{Alb})$$

*The final Lean target is `Jacobian.exists_unique_ofCurve_comp` after $D.J$ is selected as *Jacobian* C .*

Proof. Proof obligation. Milne Proposition 6.1 constructs the factor through the degree- g symmetric sum, extends the resulting rational map to the abelian target, proves it is a homomorphism by rigidity, and uses generation by g copies of the Abel image for uniqueness. The cited proposition assumes $g > 0$; the challenge theorem does not. The genus-zero terminal case and the positive-genus construction must therefore be joined explicitly. Existing rational-extension, rigidity, and conditional uniqueness files are consumers, not a theorem of type (Alb). $\square$

Theorem 37 (Phase 8 geometric and base-change assembly).

Select one original-field datum D_C for every challenge curve, with carrier definitionally equal to $\text{Jacobian}(C)$. Discharge, without changing that carrier:

1. *the group object and smooth-relative-dimension- $g(C)$ instance;*
2. *properness and geometric irreducibility;*
3. *ofCurve , its pointing law, and (Alb) ;*
4. *the contravariant functor on $\text{Curve}(K)$;*
5. *for every $K \rightarrow L$, the group-scheme base-change isomorphism, its identity and tower coherences, and compatibility with ofCurve .*

Proof. Proof obligation. Instantiate the datum-level consumers above with the same selected D_C , supply the missing geometric producers, and define every headline object by projection from that datum. Uniqueness of representers gives the base-change object isomorphism; the existing datum-level lemmas supply its group structure and Abel compatibility, after which uniqueness proves the identity and tower equations.

None of these headline targets currently has a public proof. The exact declaration-level audit, including the axiom-contaminated derived `Jacobian.congr`, is recorded in Appendix A.4; it supplies no mathematical premise to this theorem. $\square$

Appendix A

Reference map, Lean handoffs, and quarantined routes

This appendix is part of the authoritative specification. It records where each mathematical assertion comes from, what exact Lean object receives it, and which nearby constructions are deliberately outside the route.

A.1 Reference map

Step	Precise anchor	Specialization used here
R1	Kleiman, <i>The Picard Scheme</i> , Section 3, Definition 3.6, Theorem 3.7, and Exercise 3.8.	Relative effective divisors commute with base change; for a projective flat curve their functor is represented by an open Hilbert subscheme, and fixed degree is open-and-closed and of finite type. Lean proves the analogous statement only for its exact widened, locally certified degree- g functor.
R2	Kleiman, Section 3, Theorem 3.13; Milne, <i>Abelian Varieties</i> , Chapter III, Theorem 5.1(a) and Lemma 5.2(b), transcription page 0107.	An Abel fibre is a complete linear system. Under $H^1 = 0$ and rank-one pushforward, its projective bundle is the base, giving one effective divisor. The checked familywise evaluator and uniqueness theorem provide the actual Lean specialization. Milne III.5 additionally assumes complete nonsingular, positive genus, and a rational point; those are not AJCR route hypotheses.
R3	Stacks, Varieties Lemma 33.25.6, Tag 056U; Milne III, Lemma 5.2(b).	Tag 056U gives density of closed points with residue field finite separable over the base. Its use here is a derived specialization: restrict to the required nonempty smooth open, choose such a point, and then use separable closedness to split the finite separable residue extension. This is the rational-point input inside translated-drop coverage, while Milne supplies the classical degree- g , rank-one model subject to the same III.5 caveat.

Step	Precise anchor	Specialization used here
R4	Stacks, Properties of Schemes Lemma 26.15.4, Tag 01JJ; Morphisms Lemma 29.22.9, Tag 01TX; Kleiman, Theorem 4.8(1).	A sheaf covered by representable open subfunctors is represented; over the Noetherian base $\mathrm{Spec} k$, local finite type implies local finite presentation. Kleiman’s full Picard theorem is context, not an imported shortcut.
R5	Stacks, Schemes Lemmas 26.14.1–2, Tags 01JB–01JC; Limits Lemma 32.10.1, Tag 01ZM; Limits Lemmas 32.8.12 and 32.8.11, Tags 0EUU and 081E.	Descend the finite-presentation charts, maps, and equalities to a common stage; Tag 01ZM is applied to an inverse system with affine transition maps and qcqs stages, while Tags 0EUU and 081E require the relevant stage maps to be locally of finite presentation before openness or invertibility is reflected. Scheme gluing and its mapping property yield the global carrier comparison. None of these results proves Picard representability over the stage.
R6	Stacks, Fields Lemmas <i>normal-closure-inside-normal</i> , Tag 0BMG, and <i>normal-closure-galois</i> , Tag 0EXM; Picard Schemes of Curves Section 44.4, Tag 0B9K; Descent Remark 35.35.12, Tag 040L.	Take the finite normal closure of all coefficients inside the fixed normal extension Ω/K before finalizing P ; Tag 0BMG supplies containment and Tag 0EXM supplies the Galois conclusion for the finite separable input. Then descend the class and prove the all-tests Yoneda equivalence on $P.\mathrm{gluedOver}$. Tag 0B9K describes the fppf Picard sheaf only; a bespoke proof must connect it to AJCR’s big-etale degree-zero carrier. No citation supplies that connection already.
Step	Precise anchor	Specialization used here
R7	No accepted source-to-carrier specialization yet.	The one selected target is projectivity of the literal $P.\mathrm{gluedMap}$; finite hyperplane avoidance then gives orbit-affineness on $J_L = P.\mathrm{gluedOver}$. The exact conditional Lean wrapper is source-present but not public-root reachable.
R8	Stacks, Descent Section 35.38, Tag 0246, and Remark 35.35.12, Tag 040L.	Galois-stable quasi-affine charts descend effectively, and compatible morphisms descend uniquely. The checked finite-Galois quotient theorem specializes this to the represented degree-zero Picard functor.
R9	Milne III, Proposition 6.1, transcription page 0110; Kleiman, Section 5, especially Lemma 5.1 and Corollaries 5.13–5.14.	Milne proves the pointed Albanese property for positive genus; the challenge also requires genus zero. Kleiman supplies the classical group-scheme geometry. The datum-level Lean consumers retain every extra hypothesis explicitly.

A.2 Exact Lean handoffs

Step	Public handoff and exact carrier	State
R1	<code>divFunctorAff_genus_representableBy</code> on <code>divRepAffGenusScheme C</code> , representing exactly <code>divFunctorAff C</code> (genus <code>C</code>).	Proved.
R2	<code>canonicalRankOneAbelIso</code> between the two sigma-extended loci; <code>canonicalRankOneAbelSliceIso</code> slice-wise; and <code>picRankOneOpenSigmaIso</code> from the represented open <code>divRankOneOpenOver</code> .	Proved.
R3	<code>picRankOneTranslatedChart_pointwiseCover</code> for the exact translator subtype <code>PicRankOneTranslatorIndex</code> .	Proved.
R4	<code>pic0_sepClosed_representableBy</code> <code>C</code> with carrier <code>.1</code> and certificate <code>.2</code> ; the selected universal class is <code>pic0SepClosedUniversalClass C</code> .	Proved.
R5	For arbitrary legacy <code>P0 : PicOfFiniteStageGluePackage C_Omega K</code> , carrier <code>P0.gluedOver</code> ; comparison <code>P0.finiteStageBaseChangeIso</code> plus its structure-map equation.	Proved only as a carrier base-change comparison; the package theorem assumes algebraicity only and supplies no separability or Galois certificate; the convenience <code>Over</code> wrapper is unrooted.
Step	Public handoff and exact carrier	State
R6	Jointly choose legacy <code>P</code> with <code>L = P.N.1</code> finite Galois, then construct <code>xiL : pic0Subgroup ((baseChange K L).obj C) P.gluedOver, (UC)</code> , and the exact <code>RepresentableBy</code> certificate.	Unresolved; the class is the first unmet mathematical producer.
R7	Prove <code>P.gluedMap.IsProjective</code> , then construct <code>(pic0SemilinearGalActionOfRepresentableBy C repL).OrbitsInAffineOpen</code> on the same <code>P.gluedOver</code> .	Projectivity planned and blocked by R6; exact conditional wrapper source-present but unrooted.
R8	<code>pic0RepresentableBy_finiteGaloisDescent</code> <code>C repL</code> ; carrier <code>StableAffineOpen.gluedQuotientOver</code>	Proved conditionally on <code>repL</code> and orbit-affineness.
R9	<code>picRepDatum_finiteGaloisDescent</code> , then <code>PicRepDatum.toJacobianData</code> ; all shared fields preserve the quotient carrier by reflexivity.	Proved conditionally; original-field instantiation blocked by R6–R7.
P8	<code>JacobianData.ofCurve</code> , <code>pullbackHom</code> , <code>baseChange</code> , the tangent, geometry, Abel-base-change, and conditional Albanese families, then the declarations in <code>Challenge.lean</code> .	Datum consumers proved with displayed inputs; their first unmet producers and the still-sorry-bearing headline assembly are recorded below.

A.3 Quarantine

Remark 38 (Rejected and non-normative branches). The following material is not an alternate choice left to a formalizer.

1. The unrestricted high-degree or admissible Abel map, its quotient by complete linear systems, and the symmetric-power/Weil or Albanese-first constructions are rejected as construction routes. Milne’s $n > 2g$ argument cannot justify the degree- g rank-one step. The fixed admissible integer may be reused only in the downstream Phase 8 producer of an `AbelSourceData`; it does not replace R1–R8 or represent Pic^0 .
2. `DivFamZar`, the conditional F5/F6 faithfully-flat divisor classifiers, and the admissible-degree representer are implementation layers or quasi-compactness inputs only. They do not replace R1–R2.
3. A `GlueData`, `GluePackage`, overlap isomorphism, or `pic0RepresentableBy_finiteStageBaseChange` is not a representability theorem over the finite stage. Only (UC) followed by (Y) may cross R6.
4. `Pic0FiniteStageStableGluePackage` is not the R5–R7 carrier. Until a checked legacy-to-stable carrier isomorphism exists, its stable orbit wrappers cannot be spliced into this route.
5. The Lean functor `pic0TypeFunctor` means the degree-zero étale-sheafified Picard functor. No theorem currently identifies it with Kleiman’s connected identity component, so those phrases are not interchangeable in Lean attachments or completion claims.
6. The projective-space immersion entry point is only the internal proof of the selected projectivity-to-orbits implication; constructing an immersion independently is not a second R7 producer. Kleiman Theorem 4.8, arbitrary object-isomorphism transport, post-base-change representability, and filtered-colimit preservation are comparison or consumer results. None is an imported proof of R6 or R8.
7. The conditional constructors `JacobianData.ofRepresentableBy`, `ofCharts`, `ofChartsOfCompactSpace`, `ofAbelImage`, `ofAbelLifts`, `ofChartsOfAbelImage`, `ofChartsOfAbelLifts`, `ofPic0ClassSurjective`, and `PicRepDatum.toJacobianDataOfAbelLifts` are downstream packaging devices. Likewise `Pic0RepresentabilityOverlap` and `Pic0RepresentabilityDescentData` start from an already supplied representation. None replaces the quotient-carrier handoff in R8–R9.
8. `blueprint/src/chapters/detail`, `blueprint/src/chapters-legacy`, and `content-formalization.tex` are historical audit material. They are not included by `content.tex`; any conflicting route, status, or aspirational proof in them is non-normative and must not be cited as project state.

A.4 Public-root and Challenge status

The modules defining `finiteStageBaseChangeOverIso`, `pic0RepresentableBy_finiteStageBaseChange`, `pic0ClassYoneda`, `pic0SepClosedClassYoneda_app`, and `pic0FiniteStageOrbitsInAffineOpen_of_isProjective` are present in the source tree but not reachable from the public `AlgebraicJacobian` root at this checkpoint. They receive no `\leanok` credit in the normative chapters. The first two remain only slice/post-base-change comparisons, and the last remains conditional on (Proj), even after future root integration.

The headline declarations `Jacobian`, `instGrpObj`, `smoothOfRelativeDimension_genus`, the anonymous proper and geometrically-irreducible instances, `ofCurve`, `comp_ofCurve`, `exists_unique_ofCurve_comp`, the map and laws of `functor`, `baseChangeIso`, `baseChangeIso_id`, `baseChangeIso_comp`, and `baseChange_ofCurve` in `Challenge.lean` remain backed by `sorry`. `Jacobian.congr` is definitionally derived from the `sorry`-bearing functor and is therefore axiom-contaminated rather than an independent endpoint. The generic scheme base-change functor and its identity/composition isomorphisms are checked, but they do not prove the Jacobian-specific comparison or coherences.

A.5 Residual blocker ledger

- Remark 39** (Blockers in dependency order). 1. **B1, first unmet producer:** after taking the finite normal closure of all coefficients, finalize a legacy P with $L = P.N.1/K$ finite Galois, construct ξ_L on its literal `P.gluedOver`, and prove (UC), including the exact big-étale and degree-zero transports. Current tensor-stage work supplies local inputs only.
2. **B2:** prove every component of pullback by ξ_L bijective on every L -scheme, yielding (FS-Rep).
 3. **B3:** prove `P.gluedMap.IsProjective` on that exact carrier; no stable-package or independently immersed substitute is allowed.
 4. **B4:** root and instantiate the exact projectivity-to-orbits wrapper, yielding (OA) for the canonical action on the same J_L .
 5. **B5:** instantiate the conditional finite-Galois theorem and same-carrier handoff, producing one original-field `JacobianData C`.
 6. **B6:** construct the additive, scalar-compatible epsilon-kernel- $H^1(C, \mathcal{O}_C)$ comparison (SL); the current set-level equivalence cannot prove the tangent dimension formula.
 7. **B7:** prove `abelCrossBaseCechCore` for every field extension and rational base point; the current graph theorem only reduces this to two Čech-class equations.
 8. **B8:** construct the fixed proper degree- n_C Abel source, geometric reducedness, and the relative-dimension- g zero-hypercover required by the checked geometry consumers.
 9. **B9:** prove the pointed genus-zero identification with $\mathbb{P}^1$, all-test degree-zero Picard vanishing, and constancy/Hex; combine these with the positive-genus factorization to obtain (Alb).
 10. **B10:** select the datum family in `Challenge.lean`, prove functoriality and the Jacobian-specific base-change identity/tower coherences on that family, instantiate (AJ-BC), and expose all geometry without changing carriers or signatures.

B2–B10 are blocked consumers, not parallel alternatives to B1. This order is the formalization route.